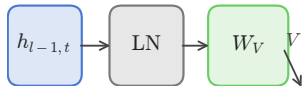
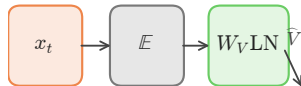


(a) V-real (ground truth)



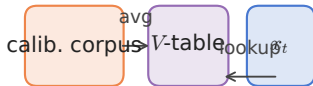
$$V = W_V \cdot \text{LN}(h_{l-1,t})$$

(b) A0 anchor (embed-proj)



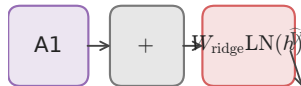
$$\hat{V}_{A0} = W_V \cdot \text{LN}(\mathbb{E}[x_t])$$

(c) A1 anchor (fitted table)



$$\hat{V}_{A1} = \mathbb{E}[V_{\text{real}} \mid \text{token} = x]$$

(d) Ridge (context residual)



$$\hat{V} = A_1 + W_{\text{ridge}} \text{LN}(h_{l-1,t})$$